

SOC/CSSS/STAT 221: Statistical Concepts and Methods for the Social Sciences

Summer 2026

Mondays and Wednesday

9:10 – 11:20 a.m.

Todd Nobles

tnobles@uw.edu

Office Hours:

Mondays 4:30 to 5:30

<https://washington.zoom.us/j/4171217812>

Or by Appointment

COURSE DESCRIPTION

This course aims for students to develop a foundation of statistical literacy and intuition. Using examples and cases from the social sciences we will cover topics such as: measurement of variables and types of data, methods to summarize numerical and categorical data; basic probability, distributions, confidence intervals, and significance tests; methods to describe relationships between variables such correlation and linear regression. We will examine how to use these techniques to answer questions about the world as well as the potential pitfalls and misuses of statistical analyses. The intuition and ability to critically analyze a piece of statistical evidence will make you a more informed consumer of the information you encounter throughout life whether that be in the form of news coverage, video shorts, or polls.

COMMUNICATION

The best way to get in touch with me with specific questions is via email (tnobles@uw.edu). However, I strongly encourage you to come to my office hours if you have larger questions or concerns about the material or the course. If you're unable to make that time, please reach out to me over email and we can arrange another time. You can expect to hear back from me within 48 hours during the week and if you email me on a Friday then you can expect a response by the end of day the following Monday.

COURSE OBJECTIVES

By the end of this course students will be able to:

- understand the meaning of basic concepts and terminology used in statistical analyses
- use descriptive and basic inferential statistics to describe relationships
- examine the strength of statistical evidence in support of a claim
- calculate basic descriptive statistics and statistical tests

COURSE MATERIALS

There are two required textbooks for this course. Both are free:

- [OpenIntro Statistics](#) 4th edition by David M. Diez, Mine Cetinkaya-Rundel, Christopher Barr. This book is available for free through the above link and has a version for screen readers as well as tablets.
- [OpenStax Introductory Statistics](#)

- Calculator for exams and in-class work

COURSE POLICIES AND EXPECTATIONS

Academic Misconduct

The University takes academic integrity very seriously. Behaving with integrity is part of our responsibility to our shared learning community. If you're uncertain about if something is academic misconduct, ask me. I am willing to discuss questions you might have.

Acts of academic misconduct may include but are not limited to:

- Cheating (working collaboratively on quizzes/exams and discussion submissions, sharing answers, and previewing quizzes/exams)
- Plagiarism (representing the work of others as your own without giving appropriate credit to the original author(s))
- Unauthorized collaboration (working with each other on assignments)

More information on what constitutes academic misconduct can be found here:

<https://www.washington.edu/cssc/for-students/academic-misconduct/>

Academic misconduct, such as the acts described above, will result in a "0" on the assignment and a referral to Community Standards and Student Conduct (CSSC). Additional offenses will result in a failing grade for the course.

Artificial Intelligence Usage

While recent developments in AI tools have brought a new set of resources for completing tasks, using these responsibly requires a baseline of knowledge and experience. This course is meant to provide that very baseline. Therefore, any use of AI tools outside of a few specifically designed instances where we may use it together or it is explicitly said to use it on an assignment is prohibited. Some amount of struggling through statistical concepts and calculations is a key part of learning. Unauthorized AI use will be considered a form of cheating and will be subject to the policy outlined under academic misconduct. If you have any questions about citation or about what constitutes academic integrity in this course or at the University of Washington, please feel free to contact me to discuss your concerns.

Classroom Conduct

A productive intellectual environment requires respectful discussion of ideas and a collaborative approach to learning. There are some basic principles of etiquette to keep in mind in a large classroom with others. A general rule of thumb before doing something you're unsure of is to ask yourself, "If the person beside me was doing XYZ, would it distract me from learning?" If the answer is yes, wait to do it until after class. Some examples of behaviors that you should avoid: showing up to class late, talking while the instructor or another student is speaking to the class, beeps/buzzes/rings from laptops or phones, computer activities that aren't course related (booking travel, online shopping, reading the news, playing video games, etc.).

ACCOMMODATIONS AND RESOURCES

Students with Disabilities

Your experience in this class is important to me. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law. If you have already established accommodations with Disability Resources for Students (DRS), please activate your accommodations via myDRS so we can discuss how they will be implemented in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), contact DRS directly to set up an Access Plan. DRS facilitates the interactive process that establishes reasonable accommodations. Contact DRS at disability.uw.edu.

Counseling Center

The University of Washington Counseling Center offers a variety of options to aid students in dealing with stress, depression, anxiety, adversity, and urgent crises. These options include individual, group, and crisis counseling, as well as workshops and resources designed to enhance student wellness. All Counseling Center services are provided to current UW students free of charge. In addition, the Counseling Center can connect students with Psychiatric Services if needed. You can schedule and view resources at wellbeing.uw.edu/unit/counseling-center/

Religious Accommodations Policy

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at <https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/>. Accommodations must be requested within the first two weeks of this course using the following form <https://registrar.washington.edu/students/religious-accommodations-request/>

COURSE ASSIGNMENTS AND GRADING

Quizzes (20%)

There will be 6 in-class quizzes throughout the quarter. These are short problem sets for demonstrating understanding of recent concepts covered in class and in the readings. The lowest score will be dropped.

Homework (35%)

Throughout the quarter we will have 6 homework assignments due. The lowest score will be dropped, so you can miss one of these with no detriment to your final grade. Each homework will be worth 7% of your final grade.

Midterm (20%)

The midterm will cover material through Week 5 of the course. It will take place on July 22nd in class.

Final Exam (30%)

There will be a final exam that will provide an opportunity to demonstrate that you've mastered the core concepts we've covered over the quarter. This will be cumulative (covering material from the full quarter) and will be during our final class meeting.

Syllabus Quiz (2% extra credit)

After the first class, there will be a syllabus quiz that you'll take on Canvas. This assignment is to ensure that you have read the syllabus and understand the course expectations and requirements.

Due Dates

Late work is accepted, however a deduction of 20% is assigned for each 24 hours after an assignment is due. However, I recognize that you all have busy lives and competing priorities with coursework, so with that in mind, you have a total of 4 "flex days" to use as you see fit to extend deadlines over the quarter. You don't need to tell me that you're using one of these days. However, you do need to keep track of these days as you use them as I won't input a final accounting of the late days until I calculate final grades at the end of the quarter. This means that you won't see a late penalty added to an assignment until later in the quarter if you've exceeded the number of allowed flex days. Additionally, if major life events occur with medical or family needs, we can work out a plan for you to complete assignments on an adjusted timeline if needed.

Final Grade Breakdown

All grades will be rounded to the nearest whole number percentage point and assigned letter grades as follows:

A+	96-100%	4.0
A	94-95%	4.0
A-	90-93%	3.7
B+	87-89%	3.4
B	84-86%	3
B-	80-83%	2.7
C+	77-79%	2.4
C	74-76%	2
C-	70-73%	1.7
D+	67-69%	1.4
D	64-66%	1
D-	60-63%	.7
F	<60%	0.0

This syllabus is subject to change at the instructor's discretion. If I make any changes, I will notify the class via Canvas and post an updated version.

Course Schedule

There are readings listed below each class session. Please complete these readings **BEFORE** coming to class that day so that we can have a productive class session. The content covered in the readings will often be needed to complete in-class exercises and to participate in discussions with your fellow students. Everyone, yourself included, will enjoy the class more if you've done the readings beforehand.

All readings below, unless noted otherwise, are from the textbook. When not from the textbook they will be posted to Canvas under Modules -> Readings.

Week	Date	Day of Week	Topic	Readings	Assignments Due
1	22-Jun-26	Monday	Introduction, Course Logistics, Levels of Measurement		
1	24-Jun-26	Wednesday	Visualizing Data and Frequency Tables	Chapter 2 OpenStax	
2	29-Jun-26	Monday	Measures of central tendency	Chapter 2 OpenStax	Homework 1 Due
2	1-Jul-26	Wednesday	Measures of variability	Chapter 3 OpenIntro	
3	6-Jul-26	Monday	Basics of probability	Chapter 3 OpenStax	Homework 2 Due
3	8-Jul-26	Wednesday	Probability and Distributions	Chapter 4 OpenStax	
4	13-Jul-26	Monday	Sampling, Basics of Inference, Confidence intervals	Chapter 8 OpenStax	Homework 3 Due
4	15-Jul-26	Wednesday	Inference for categorical variables	Chapter 6 OpenIntro	
5	20-Jul-26	Monday	Inference for numerical variables - one sample	Chapter 9 OpenStax	
5	22-Jul-26	Wednesday	Midterm		
6	27-Jul-26	Monday	Inference for numerical variables - two sample	Chapter 10 OpenStax	
6	29-Jul-26	Wednesday	ANOVA	Chapter 7 OpenIntro	Homework 4 Due
7	3-Aug-26	Monday	Correlation	Chapter 12 OpenStax	

7	5-Aug-26	Wednesday	Correlation	Chapter 12 OpenStax	Homework 5 Due
8	10-Aug-26	Monday	Bivariate regression	Chapter 8 OpenIntro	
8	12-Aug-26	Wednesday	Bivariate regression	Chapter 8 OpenIntro	Homework 6 Due
9	17-Aug-26	Monday	Multivariate Regression preview and Final exam questions and group study sessions		
9	19-Aug-26	Wednesday	Final Exam		